

CLOUDFLARE RADAR

# Cloudflare Radar 2024 Year in Review

A data-driven analysis of global Internet traffic, adoption trends, disruptions, and security threats observed in 2024.

Year in Review · 2024

# 17% Global Traffic Growth and Starlink's 3.3x Surge

Global Internet traffic growth (YoY)

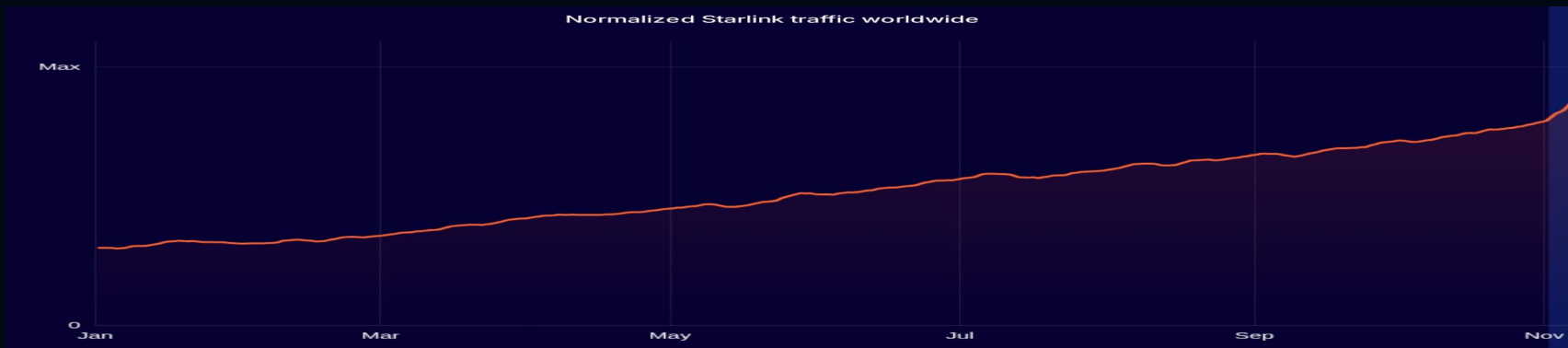
**17%**

Trend normalized from mid-January (post-holiday).

Starlink traffic growth (2024)

**3.3x**

Scaled vs. peak observed request volume.



Starlink continues expanding satellite connectivity to previously underserved locations.

# Google Leads Top 10 Services as Generative AI Rises

## Top 10 Internet services (DNS ranking)

- #1 Google
- #2 Facebook
- #3 Apple
- #4 TikTok
- #5 AWS
- #6 Microsoft
- #7 Instagram
- #8 YouTube
- #9 Amazon
- #10 WhatsApp

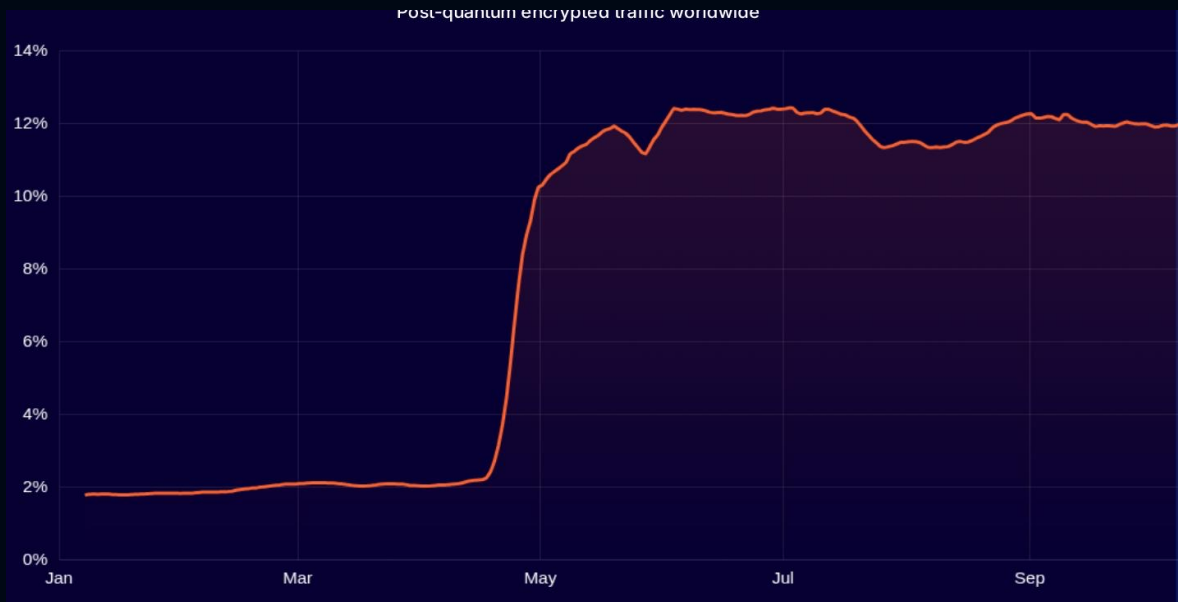
Google remains #1 for the second consecutive year (based on aggregate 1.1.1.1 resolver data).

| Rank | Service   |
|------|-----------|
| #1   | Google    |
| #2   | Facebook  |
| #3   | Apple     |
| #4   | TikTok    |
| #5   | AWS       |
| #6   | Microsoft |
| #7   | Instagram |
| #8   | YouTube   |
| #9   | Amazon    |
| #10  | WhatsApp  |

## Trend note

Generative AI continued its rapid rise throughout 2024, while the Metaverse receded into the background.

# Post-Quantum Encryption Jumps from 2% to 13% of TLS 1.3



Adoption rose sharply after Chrome 124 enabled post-quantum key agreement by default on

Post-quantum TLS 1.3

**2% → 13%**  
Approx. early 2024 to late 2024.

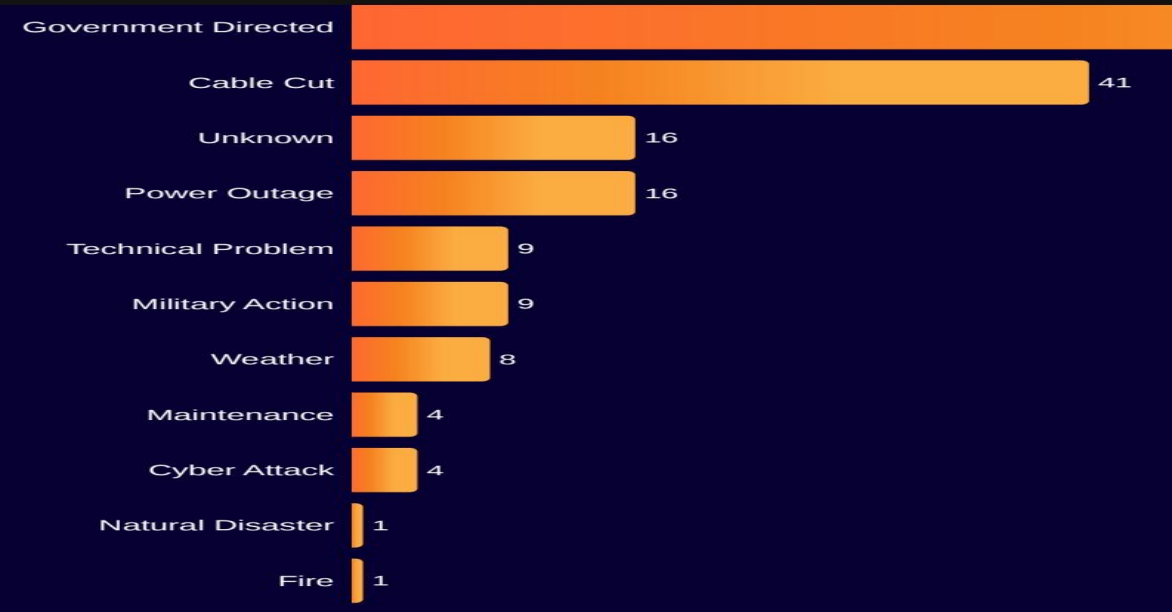
HTTP/3 adoption

**21%**  
Share of traffic using HTTP/3.

Top API client language

Also noted: Firefox rolling out post-quantum by default; Safari began initial testing.

# 225 Major Disruptions: Government Shutdowns Lead Outage Causes



By cause

|                     |      |
|---------------------|------|
| Government Directed | 116+ |
| Cable Cut           | 41   |
| Unknown             | 16   |
| Power Outage        | 16   |
| Technical Problem   | 9    |
| Military Action     | 9    |
| Weather             | 8    |
| Maintenance         | 4    |
| Cyber Attack        | 4    |
| Natural Disaster    | 1    |
| Fire                | 1    |

Government-directed shutdowns are the leading factor, followed by cable

Cloudflare observed 225 major Internet disruptions globally in 2024.

# 6.5% of Traffic Mitigated: Persistent Threats and Bot Concentration

Traffic mitigated

## 6.5%

Share of all traffic Cloudflare processed.

Most attacked sector

## Gambling / Games

Top targeted industry category.

Bot concentration

## 68%

Of global bot traffic from top 10 countries.

### Persistent vulnerabilities

Log4j (CVE-2021-44228), disclosed in December 2021, continues to be actively targeted more than two years later.

### Routing security

RPKI valid routes increased by 6.4%.

### Email security

4.3% of emails processed were malicious.

Among malicious emails, 43% contained a deceptive link.

### TLD signal

.bar originated the largest share of malicious and spam email observed during the year.

# Five Takeaways Shaping the 2025 Internet Outlook

1

## Traffic growth remains durable

Global Internet traffic grew 17% YoY in 2024; satellite connectivity (Starlink) surged 3.3×.

2

## Post-quantum momentum is real

Post-quantum encryption increased from ~2% to ~13% of TLS 1.3 traffic after Chrome 124 default enablement.

3

## Connectivity fragility is policy-driven

225 major disruptions observed; government-directed shutdowns led causes (116+), ahead of cable cuts (41) and power outages (16).

4

## Threat volume stays elevated

Cloudflare mitigated 6.5% of all traffic processed; Log4j exploitation persists years after disclosure.

2025 outlook: prioritize resilient capacity planning, protocol agility (PQC + HTTP/3), and threat-informed operations.